



Learning Sound Correspondences What About Neural Networks?

(aka a bumpy first year)

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Studying the sound evolution of languages I

Neogrammarian hypothesis: **Sound changes regularity**¹

If, at a given “point” in time, a phone in a given word changes into another phone, then all occurrences of the same phone in the same context also change in the same way.

Ex: [ka] Vulgar Latin > ... > [ʃa] French

- *cattus* *[kat.tʊs] > *chat* [ʃa] ‘cat’
- *blanca* *[blan.ka] > *blanche* [blɑ̃ʃ] ‘white (fem.)’
- *vacca* *[vak.ka] > *vache* [vaʃ] ‘white (fem.)’

Studying the sound evolution of languages II

Special words: *cognates*

Given two languages with a common ancestor, two words are cognates if they are an evolution of the same word from said ancestor, called their proto-form.

Comparative method

	A		B			C		D			E			F						
Sanskrit	y	u	g	a	m	dh	u	h	i	(tar)	s	n	u	ṣ	(ā)	-	r	u	dh	(iras)
Greek	z	u	g	o	n	th	u	g	a	(ter-)	-	n	u	-	(os)	e	r	u	th	(rós)
Latin	i	u	g	u	m	Ø	Ø	Ø	Ø	(Ø)	-	n	u	r	(us)	-	r	u	b	(er)
Gothic	j	u	k	-	-	d	au	h	-	(tar)	Ø	Ø	Ø	Ø	(Ø)	Ø	Ø	Ø	Ø	(Ø)
Gloss	'yoke'					'daughter'					'daughter-in-law'					'red'				

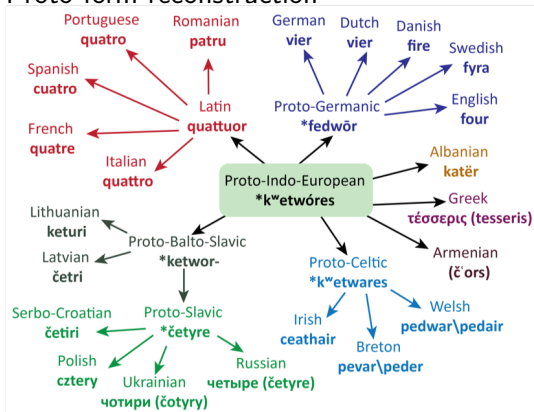
Figure 1: Regular sound correspondences across four Indo-European languages, illustrated with help of alignments along the lines of Anttila (1972: 246). In contrast to the original illustration, lost sounds are displayed with help of the dash “-” as a gap symbol, while missing words (where no reflex in Gothic or Latin could be found) are represented by the “Ø” symbol.

¹Hermann Osthoff and Karl Brugmann, *Morphologische Untersuchungen auf dem Gebiete der indogermanischen Sprachen*, volume 3 (S. Hirzel, 1880).

What are the "tasks" of phonological historical linguistics?

Cognate and sound correspondences identification

Proto-form reconstruction





How are cognate and sound change correspondences solved so far?

Automatic word comparison²

Sound alignment + distance score (Levensthein, Turchin, ...)

Sound alignment + clustering

²Johann-Mattis List, Simon J Greenhill and Russell D Gray, 'The potential of automatic word comparison for historical linguistics', *PloS one* 12:1 (2017).

³Philipp Koehn et al., *Moses: Open source toolkit for statistical machine translation*, in: *Proceedings of the 45th annual meeting of the association for computational linguistics companion volume proceedings of the demo and poster sessions* (2007).



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More recently, translation inspired methods

- Sound change prediction looks a lot like translation!

- Statistical translation methods (MOSES³)

²List et al. (as in n. 2).

³Koehn et al. (as in n. 3).

⁴Rama (as in n. 4).

⁵Dekker (as in n. 5).



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Where are the neural networks?

Siamese CNN⁴

Perceptrons and Encoder decoders⁵ : '[The] structured perceptron model outperforms the encoder-decoder RNN.' ☹

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Goal

Can we use neural networks to learn sound correspondences?
On what kind of data?

Looks like translation but small datasets, very small (and small vocabulary)

Why would we do that?

Science! Understanding! (Much debate, much linguistics)

Field linguistics⁶

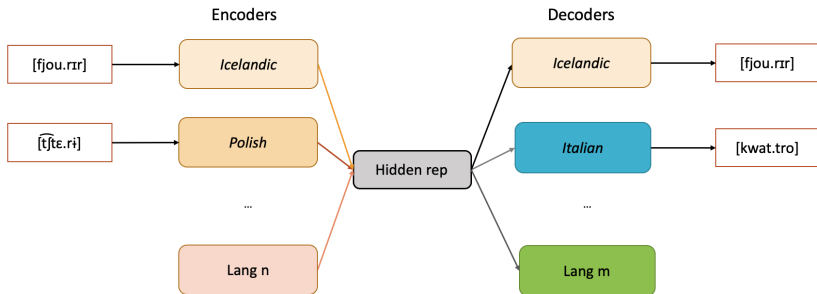


Low resource translation with transfer?⁷

⁶Timotheus A Bodt and Johann-Mattis List, 'Testing the predictive strength of the comparative method: an ongoing experiment on unattested words in Western Kho-Bwa languages', *Papers in Historical Phonology* 4 (2019).

⁷Arbi Haza Nasution, Yohei Murakami and Toru Ishida, 'A Generalized Constraint Approach to Bilingual Dictionary Induction for Low-Resource Language Families', *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* 17:2 (2017).

MEDeA - Multiway Encoder Decoder Architecture I



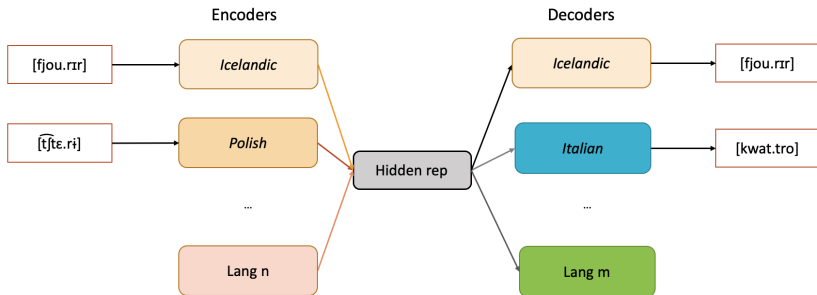
Possible model configurations

Unidirectional/Bidirectional, GRU/LSTM

Decoder with attention (Luong⁸ and Bahdanau⁹)



MEDeA - Multiway Encoder Decoder Architecture II



Losses - Accuracies

- Seq2seq (NLLLoss) - BLEU

- Siamese (Cosine Similarity) - Siamese (Cosine Embedding)

Implemented but not finished

- Transformers (will probably drop)

- Variational Encoders and Loss



Ease of use and reproducibility I

Test suite and continuous integration

Launched with JSON parameter files

```
# MODEL PARAMETERS
params_json = {
    "model_type": "CompositeModel",
    "submodels":
        [{
            "model_name": "Encoder", "lang": lang,
            "model_type": "GRU", "is_bidirectional": False,
            "num_layers": 1, "layer_dropout": layer_dropout,
            "uses_embedding_layer": True, "input_vocabulary_size": vocab_dim,
            "embedding_vector_size": emb_dim, "hidden_dimension": hidden_dim,
            "is_positive": True, "has_dense_output_layer": False
        } for lang in all_langs] +
        [{
            "model_name": "Decoder", "lang": lang,
            "model_type": "GRU", "is_bidirectional": False,
            "num_layers": 1, "layer_dropout": 0,
            "uses_embedding_layer": True, "input_vocabulary_size": vocab_dim,
            "embedding_vector_size": emb_dim, "hidden_dimension": hidden_dim,
            "output_dimension": vocab_dim, "has_dense_output_layer": True,
            "attention_type": AttentionType.LUONG_DOT
        } for lang in all_langs]
}
```



Ease of use and reproducibility II

Everything is recorded...

- Parameter files
- Data shuffling
- Model, embeddings, accuracies, loss, attention, precision, recall, ...

Through several loggers

- Console
- File
- Tensorboard

Randomness is managed (all seeds fixed)

Possibility for parallelism

Ease of use and reproducibility III

Integration with Tensorboard

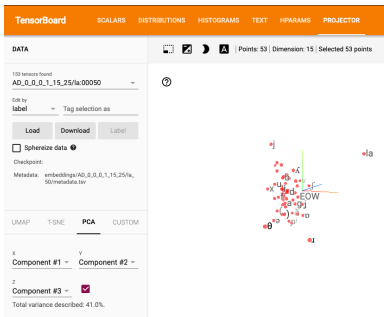


Figure 1: Embeddings

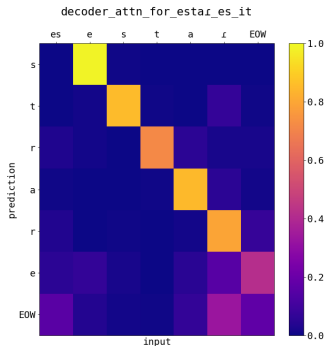
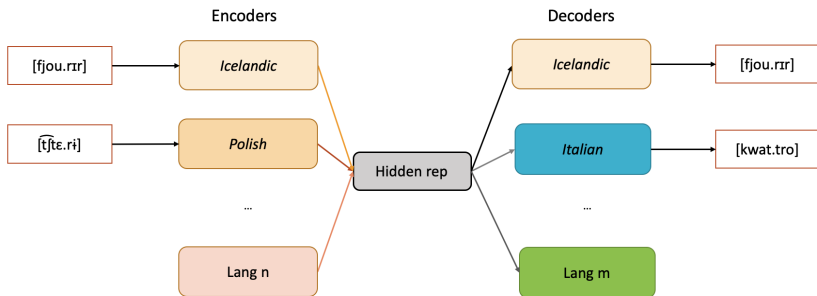


Figure 2: Attention

Reminder: This is our task



Cognate "translation" to learn sound changes

Baseline: MOSES

Accuracy: BLEU



Preliminary experiment on Balto-Slavic Languages

Data: cognates extracted from an etymological database

Dataset	PL-CZ	PL-LT	PL-IT
Cognate set	370	47	57

Data sets so small that models cannot learn

What kind of data carries the evolution of languages?



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Bilingual lexicon (train)	13,216	6,290	17,158

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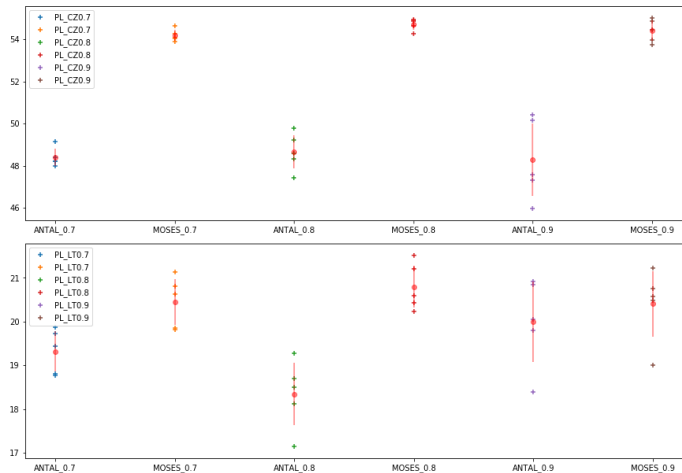


Figure 3: BLEU accuracy for PL-CZ (top) and PL-LT (bottom) cognates, for different train/validation splits and different shuffling seeds



Let's hunt problems!

Where do the learning problems come from for our NN?

Cognate datasets too small for neural networks?

Bilingual lexicons too noisy for our task?

Neural networks not adapted for this task? 🙄



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Where do the learning problems come from for our NN?

- Cognate datasets too small for neural networks?

- Bilingual lexicons too noisy for our task?

- Neural networks not adapted for this task? 🤖

What if we worked on artificial data?

- Can control size

- Can control noise level



Generating artificial data

Creating a proto-language

- Phone inventory (Romance languages)
- Phonotactics (Classical Latin¹⁰)

Creating daughter languages

- Succession of phone changes (Romance: apocope, epenthesis, palatalisation, lenition, vowel prosthesis and diphthongisation)

Results: 20,000 word triplets

[ʒolpast] > [ʒolbes], [ʒolpes]

[splutoi] > [isbledoi], [espletoi]

¹⁰András Cser, 'Aspects of the phonology and morphology of Classical Latin', Ph.D thesis, Pázmány Péter Katolikus Egyetem (2016).



Experiment

Task:

Predict sequence of phones "translation", from PL to DL, DL to DL, DL to PL

Questions:

Can MEDeA learn on a perfect (non-noisy) dataset?

How does MOSES compare to MEDeA?

What is the influence of the data size? (500, 1000, 1500, 2000, 3000)

How do the models manage ambiguity? (1, 2, 3 best - corpus BLEU)

[Zolbes] < [Zolpast]



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[Zolbes] < [Zolpast] or [Zolpest] or [Zolpas] or [Zolpes] or ...

Comparing MOSES and MEDeA on artificial data I

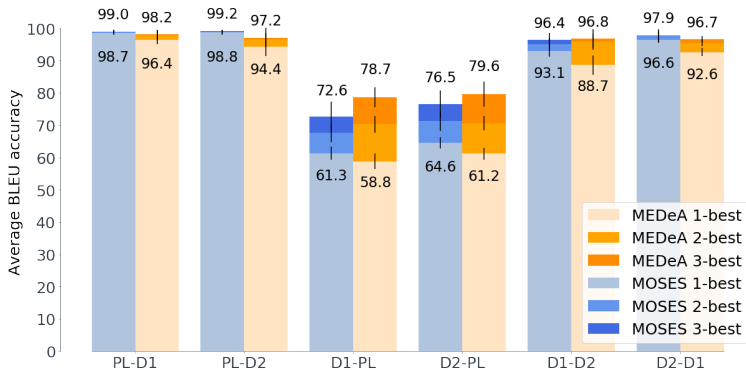


Figure 4: Comparing MOSES and MEDeA on different prediction directions

Comparing MOSES and MEDeA on artificial data II

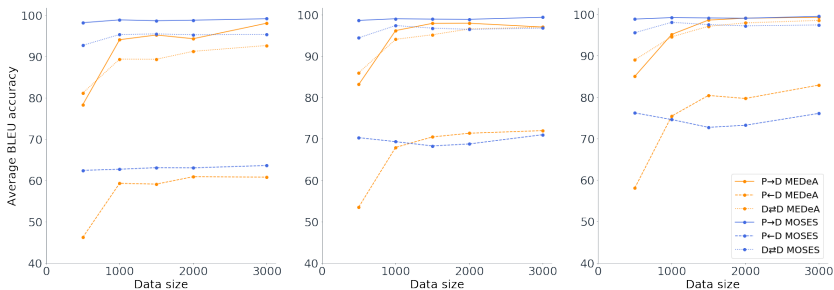


Figure 5: Influence of the data size and number of answers on the accuracy

We Need Cognates!

Database	Date	Relations	Relation types	Creation
EtymWordNet ¹¹	2013	550,000	Cognacy vs 'Other'	Wiktionary
CogNet ¹²	2019	3,000,000	Cognacy eq. to Bor!	Wiktionary Wordnet
EtymDB 1.0 ¹³	2017	500,000	Cognacy vs Bor vs Inh vs Comp vs Der	Wiktionary
COBL ¹⁴	2020 ?	?? ☹	Cognacy	Expert

¹¹Gerard de Melo, *Etymological Wordnet: Tracing The History of Words*, in: *Proceedings of the 9th Language Resources and Evaluation Conference (LREC 2014)* (2014).

¹²Khuyagbaatar Batsuren, Gabor Bella and Fausto Giunchiglia, *CogNet: A Large-Scale Cognate Database*, in: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (2019).

¹³Benoît Sagot, *Extracting an Etymological Database from Wiktionary*, in: *Electronic Lexicography in the 21st century (eLex 2017)* (2017).

¹⁴Cormac Anderson, Paul Heggarty and The CoBL Consortium, *Cognacy in Basic Lexicon database* (to be published).

Updating an Etymological Database: EtymDB 2.0

Method

2019 Wiktionary data dump > simplified xml

simplified xml > csv data base

- Word id - triplets: lexeme, gloss, language
- Relation id - Relation type, word ids

Problems with duplicate management:

Ex: παροιμίᾱ and παροιμία 'proverb'

=> grouping independently of diacritics

¹⁵Clémentine Fourrier and Benoît Sagot, *Methodological Aspects of Developing and Managing an Etymological Lexical Resource: Introducing EtymDB-2.0*, in: *Twelfth International Conference on Language Resources and Evaluation (LREC 2020)* (Marseilles, France, 2020).



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Result: Methodological paper and DB¹⁵

50% more lexemes and 40% more relations than EtymDB-1.0

But English constitutes 50% of the DB

60% of the base has a gloss (meaning)

¹⁵ibid.



Experiment

Task:

Predict sequence of phones "translation" for Romance languages - IT, ES, LA

- LA-ES-IT: 605 cognate triples (1/3 for training, 2/3 for test)
- LA-IT: 388 cognate pairs for train (extraction pb)
- LA-ES: 296 cognate pairs for train (extraction pb)
- IT-ES: 1764 cognate pairs for train
- Monolingual data: 1500 words for train

Questions:

Can MEDeA learn on a real (and noisy) dataset?

How does MOSES compare to MEDeA?

Comparing MOSES and MEDeA on real data I

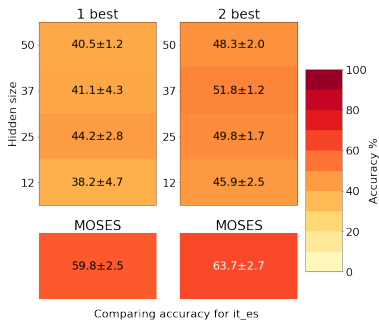


Figure 6: IT => ES

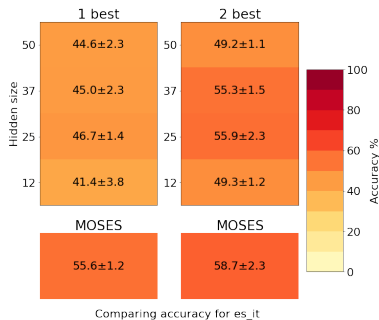


Figure 7: ES => IT

Data extraction problem: Some Spanish words appeared several times with different associations

Comparing MOSES and MEDeA on real data II

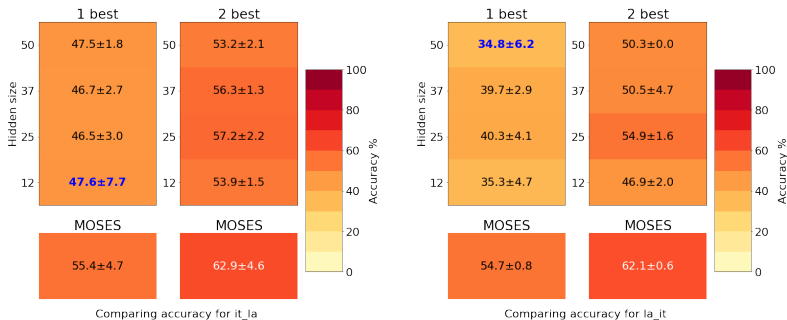


Figure 8: IT => LA

Figure 9: LA => IT

Problem with Latin: accusative or nominative



Current Status

On ideal data

- Neural networks perform better on ambiguity cases (DL=>PL)
- Statistical models perform better on small datasets

On real and very noisy data

- Data size matters, data cleanliness too, and... ?
More data extraction and cleaning need to be done
(Spanish duplicates, Latin to DL sets, Latin ambiguity)
Monolingual data needs to be used to train MOSES's LM



And afterwards?

What more can we learn from artificial data?

- Use noisy artificial data

What is the hidden representation hiding?

- Shared attentions across all decoders
- Blackbox study

Can we develop new architectures capturing sound changes better?

- Reversible networks
- Deeper architectures

[illegible]

References I

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References III

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